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In [1]:
```
from sklearn.datasets import load_breast_cancer

cancer = load_breast_cancer(as_frame=True)
```

In [2]:
```
cancer.data.head(20)
```

Out[2]:

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points
0	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.30010	0.14710
1	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.08690	0.07017
2	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.19740	0.12790
3	11.42	20.38	77.58	386.1	0.14250	0.28390	0.24140	0.10520
4	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.19800	0.10430
5	12.45	15.70	82.57	477.1	0.12780	0.17000	0.15780	0.08089
6	18.25	19.98	119.60	1040.0	0.09463	0.10900	0.11270	0.07400
7	13.71	20.83	90.20	577.9	0.11890	0.16450	0.09366	0.05985
8	13.00	21.82	87.50	519.8	0.12730	0.19320	0.18590	0.09353
9	12.46	24.04	83.97	475.9	0.11860	0.23960	0.22730	0.08543
10	16.02	23.24	102.70	797.8	0.08206	0.06669	0.03299	0.03323
11	15.78	17.89	103.60	781.0	0.09710	0.12920	0.09954	0.06606
12	19.17	24.80	132.40	1123.0	0.09740	0.24580	0.20650	0.11180
13	15.85	23.95	103.70	782.7	0.08401	0.10020	0.09938	0.05364
14	13.73	22.61	93.60	578.3	0.11310	0.22930	0.21280	0.08025
15	14.54	27.54	96.73	658.8	0.11390	0.15950	0.16390	0.07364
16	14.68	20.13	94.74	684.5	0.09867	0.07200	0.07395	0.05259
17	16.13	20.68	108.10	798.8	0.11700	0.20220	0.17220	0.10280
18	19.81	22.15	130.00	1260.0	0.09831	0.10270	0.14790	0.09498
19	13.54	14.36	87.46	566.3	0.09779	0.08129	0.06664	0.04781

20 rows × 30 columns

In [3]:
```
cancer.target.value_counts()
```

Out[3]:
```
target
1    357
0    212
Name: count, dtype: int64
```

3. Tách dữ liệu thành hai phần: training data(70%) và test data (30%

In [4]:
```
from sklearn.model_selection import train_test_split
train_x , test_x, train_y, test_y  = train_test_split(cancer.data, cancer.ta
```

4. Chuẩn hóa dữ liệu với StandardScaler

In [5]:
```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
fitted_train_x = scaler.fit_transform(train_x)
fitted_test_x = scaler.fit_transform(test_x)
```

5. Tạo và huấn luyện mô hình với training data

In [6]:
```
from sklearn.tree import DecisionTreeClassifier
tree_clf = DecisionTreeClassifier(max_depth=2)
tree_clf.fit(fitted_train_x, train_y)
```

Out[6]:

▾ DecisionTreeClassifier

▶ Parameters

In [7]:
```
!dot -Tpng cancer.dot -o cancer.png
```

In [8]:
```
pred = tree_clf.predict(fitted_test_x)
```

In [13]:
```
from sklearn.metrics import classification_report, confusion_matrix
print(classification_report(test_y, pred))
```

```
              precision    recall  f1-score   support

           0       0.98      0.77      0.86        70
           1       0.86      0.99      0.92       101

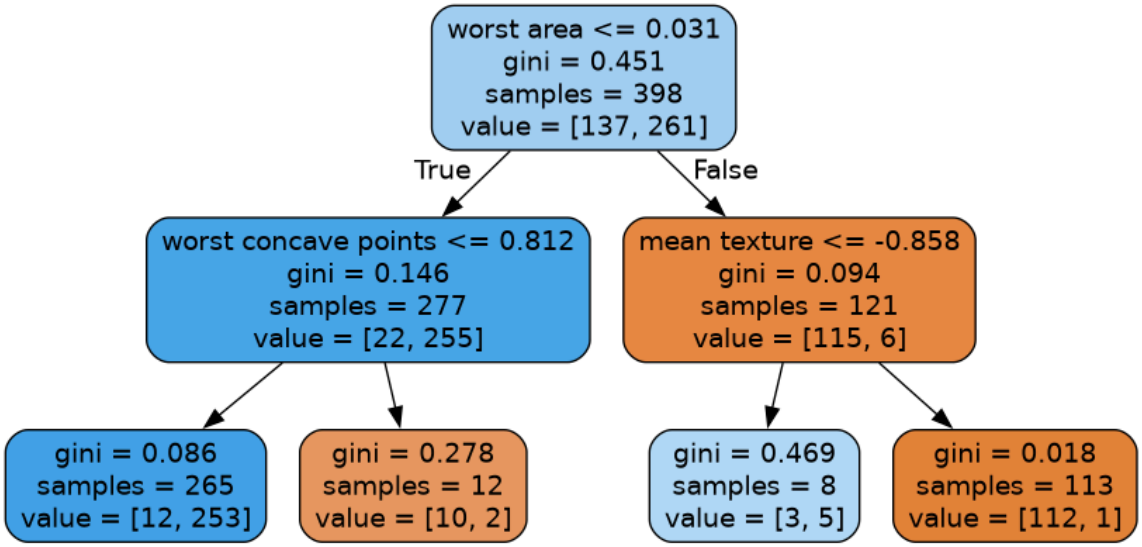
    accuracy                           0.90       171
   macro avg       0.92      0.88      0.89       171
weighted avg       0.91      0.90      0.90       171
```

In [12]:
```
confusion_matrix(test_y, pred)
```

Out[12]:
```
array([[ 54,  16],
       [  1, 100]])
```

In [11]:
```
from sklearn.tree import export_graphviz
export_graphviz(
    tree_clf,
    out_file="cancer.dot",
```

```
    feature_names=train_x.columns,
    rounded=True,
    filled=True
)
```



In [ ]: